Experiment 4: Data Modelling and Form Design

1. AIM

To design a data model and create forms using Zoho Creator by identifying suitable data fields, selecting appropriate field types, configuring field properties, and establishing relationships between application data.

2. THEORY

Data Modelling is the process of organizing data into entities, attributes, and relationships to ensure efficient storage, retrieval, and management of information. A well-designed data model reduces data duplication, improves data consistency, and supports efficient application development.

In low-code development, data modelling is simplified through visual form builders and predefined field types. Zoho Creator stores submitted form data as application records and provides tools to organize, validate, and retrieve that data.

Form Design is the process of creating user interfaces for collecting data. Forms contain fields such as text boxes, numbers, email addresses, dates, dropdowns, radio buttons, and checkboxes. Field properties can be configured to make fields mandatory, validate input, and improve data accuracy.

The major components of data modelling and form design are:

- **Entity:** An object about which data is stored, such as a student, product, or service request.
- **Attribute:** A property of an entity, such as name, email, or date.
- **Field Type:** Defines the kind of information a field can store.
- **Validation:** Ensures that entered data follows specified rules.
- **Relationship:** Represents an association between records in different forms.
- **Form Layout:** Organizes fields into logical sections for convenient data entry.

Zoho Creator provides a drag-and-drop form builder that enables developers to create forms with minimal coding.

3. PROCEDURE

1. Log in to the Zoho Creator platform.
2. Open an existing application or create a new application.
3. Navigate to the application builder and select the option to create a new form.

4. Enter an appropriate form name.
5. Open the form builder and examine the available field types.
6. Drag and drop the required fields into the form.
7. Configure each field's display name, field name, and properties.
8. Select suitable field types according to the data being collected.
9. Mark important fields as mandatory.
10. Configure validation rules for email addresses, phone numbers, numeric values, and dates where applicable.
11. Organize related fields into sections to improve readability.
12. Establish lookup relationships between forms if the application requires shared data.
13. Save the form and preview it to verify the layout and field configuration.
14. Submit sample records to check that the form stores data correctly.

4. DATA MODELLING AND FORM DESIGN EXAMPLE

A Vehicle Service Request Form can be designed using the following fields:

Field Name	Field Type	Purpose
Owner_Name	Name	Stores the vehicle owner's name
Email_Address	Email	Stores the owner's email address
Contact_Number	Phone	Stores the contact number
Vehicle_Registration_Number	Single Line	Stores the vehicle registration number
Vehicle_Type	Dropdown	Selects the vehicle category
Odometer_Reading	Number	Stores the current odometer reading
Service_Type	Radio	Selects the required service category
Issue_Description	Multi Line	Describes the vehicle problem
Preferred_Service_Date	Date	Stores the preferred service date
Drop_Off_Time	Time	Stores the preferred drop-off time
Pickup_Address	Address	Stores the pickup or drop-off address
Authorized_Estimate_And_Repair	Checkbox	Records authorization to proceed

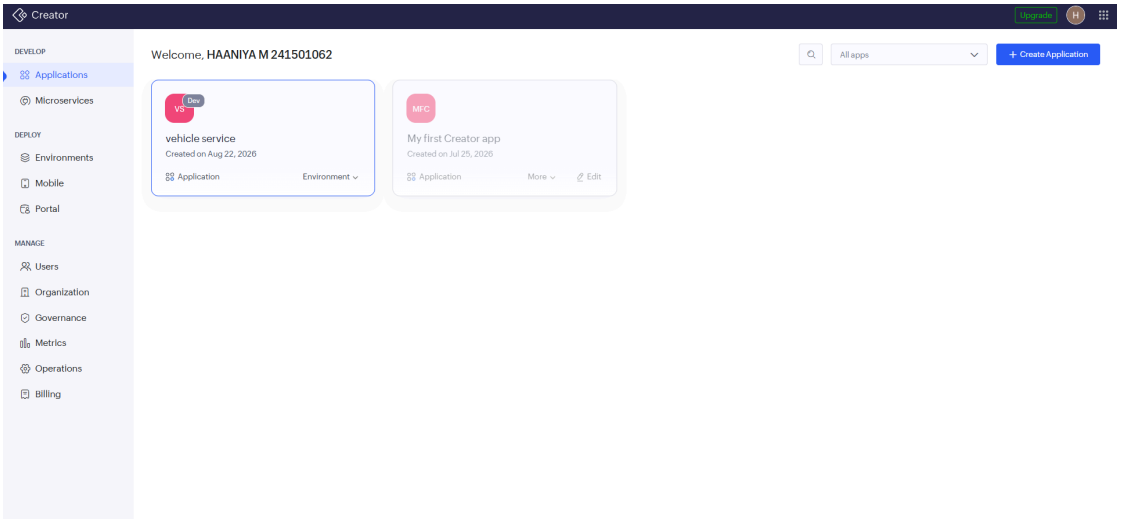
Form Design Principles

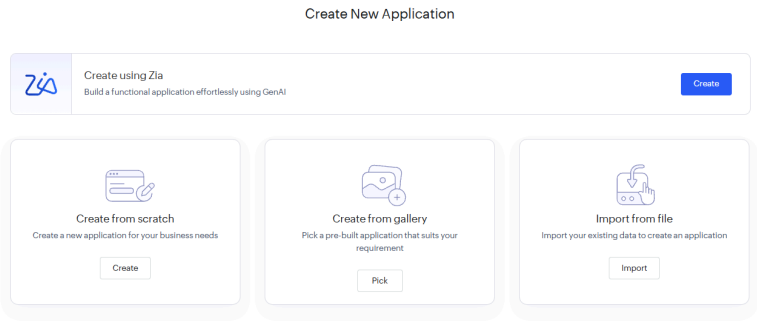
- Use suitable field types for each data item.

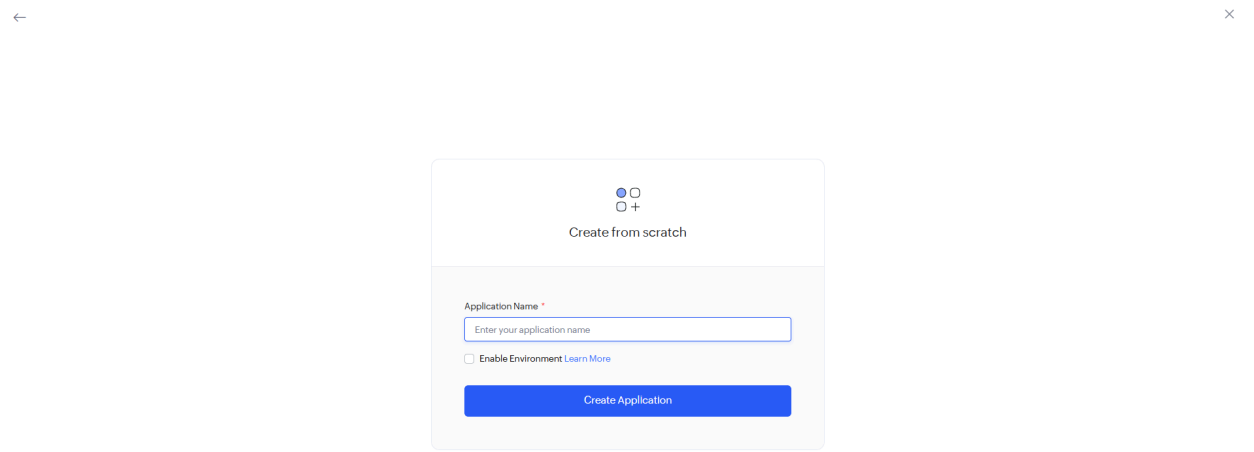
- Make essential fields mandatory.
- Apply validation to prevent invalid entries.
- Group related fields into sections such as Owner Details, Contact Details, Vehicle Details, and Scheduling.
- Use dropdowns and radio buttons when users must select from predefined choices.
- Use lookup fields when data needs to be referenced from another form.
- Maintain a clear and consistent layout.

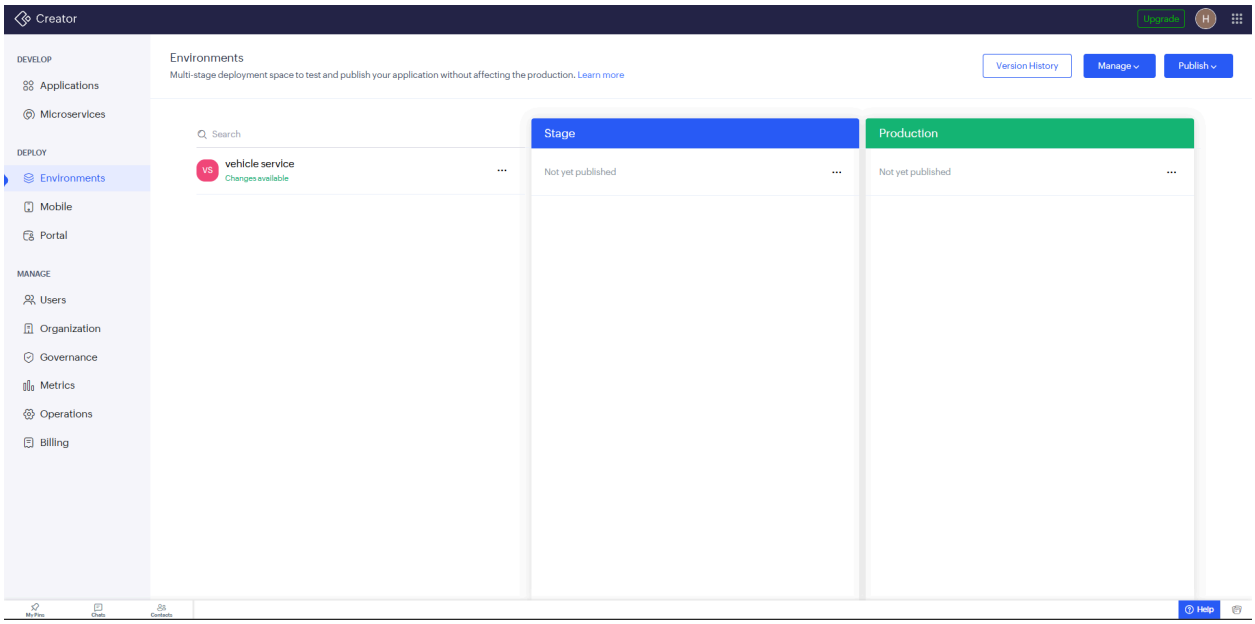
5. RESULT

Data modelling and form design were successfully studied using Zoho Creator. A form was designed using appropriate field types, field properties, validation rules, and logical sections. The experiment demonstrated how visual form builders simplify data collection and how a structured data model improves data consistency, organization, and application usability.






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